CSE4203 Principles of Software Architecture

RapidMed

Emergency Triage and Real-Time Ambulance Dispatch System

Milestone 1: Proposal and architectural requirements

Task 2: Stakeholders and concerns

By Omar Aladham

Team	RapidMed
Course	CSE4203 Principles of Software Architecture
Instructor	Dr Nada Mahmoud Hassan
Teaching assistant	Eng. Mohamed Fareh
Milestone	Milestone 1, proposal and architectural requirements
Date	23 October 2026

Team members

Name	Student ID
Omar Aladham	231015308
Marwan Abd Elmoneim	231008132
Ziad Ehab	231014691

College of Computing and Information Technology
Arab Academy for Science, Technology and Maritime Transport

2. Stakeholders and concerns

Stakeholder	Key concerns
Field paramedics End users: app on the ambulance's computer	• The app keeps working with a weak or no signal and never loses entered data • Recording vital signs takes a few taps, even with gloves on • A destination within seconds of asking • The patient's allergies, conditions and medications on screen quickly, even when the patient cannot speak
Hospital triage coordinators End users: web portal	• Two critical patients are never sent for one remaining bed • Crews see the capacity the coordinator set, and how old it is • Early notice of each incoming patient, with vital signs and arrival time • Updating capacity takes seconds during a busy shift
Central dispatch supervisors End users: web portal	• A live, accurate view of every ambulance and hospital • Overrides reach the crew at once and are confirmed • Alerts when a crew loses its connection or a hospital fills • The reason behind each recommendation can be seen
Patients and their families Data subjects, not users	• Taken without delay to a hospital that can treat them • Medical history seen only by those treating them • Every emergency access recorded and open to review
Ambulance service and hospital management Business owners	• Fewer wasted trips and shorter handovers • Reports on transport times, reservations and overrides • Little extra work for crews and hospital staff • Low running costs for hosting and API calls
Personal Data Protection Center Regulator, Egypt	• Health data are sensitive personal data under Law No. 151 of 2020 • Access only by those who need it, for a stated purpose • Data protected in storage and in transit • Breaches reported to the Center within 72 hours • Break-glass use justified, logged and reviewed
Sanad PHDMS owners Owners of the patient health data management system that RapidMed reads from	• Only registered systems and authorised staff can query records • Every query traceable to a person and a reason • Standard FHIR requests, within agreed rate limits
Operations staff Operators	• Each of the three parts deployed and rolled back on its own • Monitoring and alerts for each part and each external service • API keys and other secrets kept out of the code • Graceful behaviour when an external service fails
Development team Developers (this team)	• Clear boundaries, so that three people can work in parallel • Every external service replaceable by a mock for testing • Free service tiers enough for the prototype • Milestone deadlines met